SLIIT UNI
THE KNOWLEDGE UNIVERSITY

**IT4010 Research Project – 2025 July**

**Topic Assessment Form**

**Project ID:** 25-26J-305

1. Topic (12 words max)

Smart Ayurvedic Disease Prediction and Recommendation System.

2. Research group the project belongs to

**CoEAI - Centre of Excellence for AI**

3. Specialization of the project belongs to

**Information Technology (IT)**

4. If a continuation of a previous project:

| | |
|---|---|
| Project ID | |
| Year | |

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Ayurveda is one of the oldest systems of medicine, focusing on treating the root causes of diseases through natural remedies, lifestyle changes, and diet to maintain health and prevent illness [1]. It is widely respected for its ability to manage mild conditions such as headaches, digestion problems, colds, and stress, while also promoting long-term health and balance. In Sri Lanka, Ayurveda plays an important role in traditional healthcare, and many people still prefer natural healing methods over chemical-based treatments. Even though Ayurvedic treatments are known to be effective, especially for prevention and daily health maintenance, one of the main challenges is the limited accessibility to qualified Ayurvedic practitioners and personalized guidance. In many parts of the country, it can be difficult for people to meet a trained Ayurvedic doctor. As a result, individuals often rely on general online content, social media posts, or self-medication without fully understanding their body constitution or the root cause of their symptoms. Another concern is that, unlike modern healthcare systems which now use advanced technologies like artificial intelligence and data analysis, Ayurveda in Sri Lanka has not yet fully adopted such digital tools. There are very few smart, user-friendly systems available to help users identify possible health issues or get basic guidance based on Ayurvedic principles. This creates a gap between the growing public interest in natural medicine and the tools available to support that interest in a reliable and educational way

[1] H. Sharma, H. M. Chandola, G. Singh, and G. Basisht, "Utilization of Ayurveda in health care: An approach for prevention, health promotion, and treatment of disease," *J. Altern. Complement. Med.*, vol. 13, no. 10, pp. 1135–1

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

To address the accessibility and personalization gap in Ayurvedic healthcare, we propose an AI-based Ayurvedic disease prediction and recommendation system. This system is designed to provide users with personalized diet recommendations, Ayurvedic medicine or herbal remedies, and an intelligent Question-Answering (QA) agent for further Advice Users will be able to input their health concerns such as symptoms and basic information. The system will analyze these inputs using trained machine learning models and predict possible mild health conditions based on Ayurvedic principles. Once a condition is identified, the system will recommend suitable Ayurvedic treatments and lifestyle changes, including personalized diet suggestions. For users seeking more information, the built-in QA agent will provide interactive, natural language responses. This module is designed to answer questions related to the user's health condition, Ayurvedic concepts. The primary target audience includes individuals who lack easy access to Ayurvedic practitioners. Especially in rural areas or during situations where in-person consultations are not feasible. This solution encourages self-care and preventive health practices while minimizing the risks associated with self-medication and misinformation. The system is designed to be accessible via web or mobile platforms and features a user-friendly interface. Explanations for each recommendation are also provided to ensure users understand the reasoning behind the system's suggestions, promoting transparency and trust in the platform.

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

**Domain Expertise Required:** The project demands comprehensive understanding of Ayurvedic diagnostic principles, particularly the tri-dosha theory (Vata, Pitta, Kapha) which forms the foundation of disease classification and treatment in Ayurveda. Expertise in constitutional analysis (Prakriti assessment) is essential, as individual body types significantly influence disease manifestation and symptom presentation and prediction. Deep knowledge of classical Ayurvedic texts (Charaka Samhita, Madhava Nidana, Sushruta Samhita) for accurate symptom-disease relationship extraction is crucial.

Technical expertise includes machine learning for healthcare applications, natural language processing for Sanskrit-English medical terminology, clinical validation methodologies for traditional medicine systems, and fuzzy logic systems for uncertainty handling.

**Knowledge Requirements:** Understanding of dosha imbalance patterns and their correlation with specific diseases is fundamental. Knowledge of Ayurvedic symptom interpretation, where identical symptoms may indicate different conditions based on patient constitution, is critical for accurate model development. Familiarity with traditional diagnostic methods (pulse analysis, tongue examination) and their computational representation is necessary.

**Data Requirements:**

- **Classical Text Data**: Digitized symptom-disease mappings from authenticated Ayurvedic literature
- **Constitutional Assessment Data**: Prakriti questionnaires and scoring mechanisms validated by Ayurvedic practitioners
- **Patient Symptom Records**: Structured datasets containing symptoms with constitutional type annotations
- **Expert Validation Data**: Practitioner diagnoses for model validation and comparison
- **Terminology Datasets**: Sanskrit-English medical term mappings with contextual meanings
- **Demographic Data**: Age, gender, geographic factors affecting constitutional variations
- **Treatment Protocols**: Medicine-disease mappings with dosage and preparation methods

The integration requires careful preservation of authentic medical principles while ensuring computational feasibility and clinical validation through collaboration with qualified Ayurvedic practitioners.

8. Objectives and Novelty

| Main Objective<br>Develop an integrated AI System for Ayurvedic healthcare that combines traditional principles with modern ML techniques for personalized disease prediction and treatment and medicine recommendation along with diet recommendation and intelligent question answering system over ayurvedic medicine. | | | |
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| Member Name with Registration No | Sub Objective | Tasks | Novelty |
| Roche J P<br>IT22344274 | Medicine Recommendation Module | • Extract disease–herb relationships from Ayurvedic sources<br>• Apply Named Entity Recognition (NER**)** to identify and label relevant entities<br>• construct a labeled dataset of disease and corresponding Ayurvedic medicines<br>• Perform feature engineering by incorporating predicted encoding relevant attributes like dosha types,user profile, and predicted disease labels | • context-aware Ayurvedic medicine recommendation system that personalizes suggestions based on disease<br>• Structured Ayurvedic dataset creation |

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| | | • Implement a classification model for medicine recommendation<br>• Build a web interface for medicine recommendation input and output, integrating predicted disease<br>• Compare performance across models | |
| Perera S I A<br>IT22905918 | Ayurvedic Disease Prediction Module | • Extract symptom-disease relationships from classical Ayurvedic texts (Charaka Samhita, Madhava Nidana)<br>• Develop NER models for Sanskrit medical terminology extraction.<br>• Collect symptom records through surveys<br>• Implement baseline classification models (Random Forest, SVM, Neural Network)<br>• Develop constitutional weighting mechanism for top 10 diseases<br>• Create symptom severity scoring (3-level: mild/moderate/severe) | • This will be the first prototype incorporating basic Prakriti assessment in ML disease prediction.<br>• Comparative study showing constitutional features' impact on prediction accuracy<br>• Small-scale annotated Ayurvedic medical dataset<br>• Baseline methodology for future comprehensive systems |

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| | | • Build web interface for symptom input and prediction<br>• Validate with 2-3 Ayurvedic practitioners<br>• Compare performance with/without constitutional features and document results and write thesis | |
| Fernando K P M R A<br>IT22897176 | Intelligent Question Answering (QA) over ayurvedic medicine/advice module | • Collect and preprocess Ayurvedic texts and FAQs to form a domain-specific corpus for question answering.<br>• Implement Named Entity Recognition (NER) to extract key Ayurvedic concepts.<br>• Develop or compile a structured question–answer dataset comprising contextual passages, user queries, and corresponding answers grounded in authentic Ayurvedic knowledge sources (Charaka Samhita, Madhava Nidana).<br>• Integrate a retrieval-based approach using sentence embeddings and a vector | • Domain-specific QA system for Ayurveda: Tailored question answering in a traditional medical domain where very limited structured NLP work exists.<br>• Hybrid QA architecture: Combines retrieval-based techniques with fine-tuned transformers for high accuracy and explainability.<br>• Custom Ayurvedic QA dataset creation: First-of-its-kind structured dataset for question-answering using authentic Ayurvedic texts and manuals.<br>• Understand complex Ayurvedic questions by identifying key terms like |

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| | | database to fetch relevant content from the Ayurvedic corpus.<br>• Fine-tune QA models (BERT, T5) to generate accurate, context-aware answers from retrieved content.<br>• Evaluate the QA system using NLP evaluation metrics (F1, BLEU) and human expert validation for domain relevance. | doshas, remedies, and treatments using Named Entity Recognition (NER) and word embeddings. |
| DIAS W A N M<br>IT22899910 | Ayurvedic Dietary Recommendation Module | • Extract dosha–food relationships from classical Ayurvedic texts and validated Ayurvedic diet manuals.<br>• Develop food classification rules based on ayurveda for a dataset<br>• Design a rule-based engine to generate daily/weekly diet plans.<br>• Use simple logic + lookup tables to personalize food suggestions. | • Context-aware Ayurvedic dietary recommendation system that personalizes meal plans based on disease, dosha, and seasonal factors<br>• Structured Ayurvedic food dataset creation<br>• Integrated rule-based and feature-driven approach to align classical Ayurvedic diet principles with modern personalization needs |

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| | | • Compare model-based performance on accuracy, user satisfaction.<br><br>• Incorporate adaptive logic to adjust meals based on digestion, mood, sleep, age and all possible attributes(user-rated).<br><br>• Consult 1–2 Ayurvedic dieticians or practitioners to verify food-dosha mappings. | |

9. Individual component description of how it is complied with the specialization.

<table>
<tr><th>Member Name with Registration No</th><th>Description</th></tr>
<tr><td>Roche J P<br>IT22344274</td><td>The Ayurvedic Medicine Recommendation module supports users by providing tailored medicinal suggestions based on disease prediction, symptom intensity, age, gender, and individual constitution. It combines intelligent data handling, logical decision-making, and user-focused design<br>Personalized Recommendations:<br>Suggests Ayurvedic medicines based on disease prediction, symptom severity, age, gender, and user-specific body constitution.<br>Supervised Learning: Machine learning algorithms are trained on labeled data to recommend one or more Ayurvedic medicines based on input features. This supports intelligent, data-driven decision-making.<br>Contextual Feature Integration: Use of dosha types, symptom levels, and demographic information to enhance personalization within the recommendation logic.<br>Model Optimization: Comparison of rule-based and data-driven models using evaluation metrics such as Hamming loss, accuracy, and F1-score to determine the most effective approach.<br>Integrated IT Design:<br>Merges structured data handling, logical inference, and user-centered interface development to ensure a responsive and intelligent experience.<br>Curated Knowledge Base:<br>Maintains a structured database linking diseases with Ayurvedic herbs, compiled from classical texts and expert validation.</td></tr>
<tr><td>Perera S I A<br>IT22905918</td><td>The disease prediction component directly applies core AI/ML principles through supervised learning algorithms, feature engineering, and model evaluation techniques. Specialization alignment includes<ul><li>Machine Learning: Implementation of classification algorithms (Random Forest, SVM, Neural Networks) for multi-label disease prediction</li><li>Deep Learning: Neural network architectures for complex symptom pattern recognition</li><li>Feature Engineering: Novel constitutional-based feature creation and symptom weighting mechanisms</li><li>Model Evaluation: Cross-validation, performance metrics, and comparative analysis methodologies</li><li>Uncertainty Quantification: Probabilistic models for handling diagnostic uncertainty inherent in traditional medicine.</li></ul></td></tr>
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| | This component demonstrates advanced AI/ML concepts while addressing real-world healthcare challenges, showcasing the application of computational intelligence in traditional medicine domains. |
| Fernando K P M R A<br>IT22897176 | The Intelligent Question Answering (QA) system integrates advanced Information Technology concepts such as **Natural Language Processing (NLP)**, **Machine Learning (ML)**, and **Information Retrieval** to provide users with accurate, automated answers to queries related to Ayurvedic medicine. This component directly aligns with the IT specialization by applying core IT skills in the development of intelligent systems, particularly in the areas of **data processing, AI-based search systems, knowledge representation**, and **human-computer interaction**.<br><br>By building a domain-specific QA system, the project emphasizes:<br><br>• **Text processing and semantic understanding** using NLP<br>• **Model training and fine-tuning** using machine learning algorithms<br>• **Knowledge base construction** and embedding techniques for Ayurvedic content<br>• **Retrieval-augmented generation** for dynamic user interaction<br>• **Deployment of smart systems** that behave like human-like understanding<br><br>This component showcases the practical application of IT skills in real-world domains such as **healthcare and traditional medicine**, demonstrating how modern technologies can be used to digitize and make accessible ancient knowledge systems like Ayurveda. |
| Dias W A N M<br>IT22899910 | The Dietary Planning Module applies IT concepts such as data modeling, logic-based systems, and user-centered web interface design to generate personalized Ayurvedic diet plans. It involves constructing a structured dataset of food items with Ayurvedic properties, implementing rule-based dietary logic, and integrating contextual user data (e.g., dosha, disease, season) for tailored recommendations.<br><br>**Knowledge Representation**: Structured dataset creation for Ayurvedic food items annotated with Rasa, Virya, Vipaka, and dosha impact<br>**Rule-Based Reasoning**: Development of an expert system that maps user input into dietary suggestions using domain-specific logic<br>**Feature Engineering:** Incorporation of user dosha type, disease condition, and seasonal context as input variables for diet generation |

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| | **Personalization Logic**: Scoring mechanisms to prioritize foods based on suitability, compatibility, and dietary restrictions<br>**ML Integration (Optional Extension):** Recommendation learning via feedback-based adaptation and preference modeling for future enhancement |

10. Supervisor details

| | Title | First Name | Last Name | Signature |
|---|---|---|---|---|
| Supervisor | | | | |
| Co-Supervisor | | | | |
| External Supervisor | | | | |
| Summary of external supervisor's (if any) experience and expertise | | | | |

**<u>This part is to be filled by the Topic Screening Staff members.</u>**

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

| Yes | | No | |
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b) Does the proposed topic exhibit novelty?

| Yes | | No | |
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c) Do you believe they have the capability to successfully execute the proposed project?

| Yes | | No | |
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d) Do the proposed sub-objectives reflect the students' areas of specialization?

| Yes | | No | |
|---|---|---|---|

e) Supervisor's Evaluation and Recommendation for the Research topic:

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**IT4010 Research Project – 2025 July**

**Topic Assessment Form**

Acceptable: Mark/Select as necessary

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| Topic Assessment Accepted | |
| Topic Assessment Accepted with minor changes* | |
| Topic Assessment to be Resubmitted with major changes* | |
| Topic Assessment Rejected. Topic must be changed | |

* Detailed comments given below

Comments

| Staff Member's Name | Signature |
|---|---|
| | |
| | |

**Important*:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.